

SIGNAL-TO-CONSTRUCT THRESHOLDS · EVIDENCE REFERENCE

Numerical bands — and where they actually hold.

Low / Medium / High thresholds for biometric & behavioural signals, from the peer-reviewed literature — with the boundary between absolute cutoffs and baseline-relative bands drawn explicitly.

Two kinds of number live here. (A) **Absolute reference cutoffs** — published and quasi-standard. Only two signals qualify: **PERCLOS** (drowsiness) and the **PVT response-latency lapse** (attention). (B) **Baseline-relative bands** — everything else. Resting pupil size, blink rate, fixation duration and speech rate vary enormously between people and with task and lighting, so the literature gives a reference range, a direction, and an effect magnitude — which become Low / Medium / High **relative to each person's own calibrated baseline**, never a population cutoff.

No biometric thresholds exist for **Stress Response** (weak / contested) or **Challenge-Skill Balance** (no biosignal correlate at all). Those are not given numbers here, by design.

THE BASELINE-RELATIVE BANDING CONVENTION

engineering convention — not a published cutoff

LOW / normal

within ± 1 SD of the individual's baseline ($\approx \pm 10$ – 15 %).

MEDIUM / elevated

1–2 SD from baseline (≈ 15 – 30 % change), sustained over the window.

HIGH

> 2 SD from baseline ($\approx > 30$ %), sustained and corroborated by ≥ 1 other signal.

01 COGNITIVE LOAD

pupil · blink · gaze · speech timing

SIGNAL	WHAT THE SCIENCE SAYS (REFERENCE · DIRECTION · MAGNITUDE)	LOW / MEDIUM / HIGH	PRINCIPAL CAVEAT
Pupil dilation (TEPR) STRONG	Resting 2–8 mm (light-dependent). Cognitive load → dilation of +0.1 to +0.5 mm above pre-task baseline, scaling with difficulty.	L $\leq +0.1$ mm M $+0.1$ – 0.3 mm H $> +0.3$ mm vs baseline	Dominated by the pupillary light reflex — needs luminance control or a luminance-independent index. Webcam $\ll$ infrared.
Blink rate — visual task MODERATE	Baseline ~ 15 – 20 /min (range 10–25). Visual load → suppression , can fall to $\sim \frac{1}{2}$ of baseline.	L within ± 15 % M 15–40 % below H > 40 % below baseline	Direction reverses for non-visual load (blinks $\uparrow$). Raw count is ambiguous without task + baseline.
Fixation duration MODERATE	~ 200 ms reading · ~ 330 – 350 ms scene. Load → longer fixations ; > 500 ms during difficult integration.	L near task baseline M $+20$ – 50 % H $> +50$ % or recurrent > 500 ms	Highly task/content-dependent; meaningful only vs a defined task and the person's own scanning baseline.
Fixation count, scan-path, dwell MODERATE	Task-specific. Load → more fixations & regressions , shorter saccades, longer/disrupted scan-path, longer AOI dwell. "Irregularity, not uniform slowing."	L near task baseline M 1–2 SD above H > 2 SD above baseline	Descriptive, not a load index alone; needs AOIs; dwell reflects interest or difficulty.
Speech rate & pauses MODERATE	Articulation ~ 5 syll/s; conv. ~ 120 – 150 wpm; silent-pause cutoff 250 ms. Load → rate $\downarrow$, pauses & filled-pauses $\uparrow$.	L ± 10 % of baseline rate M 10–25 % slower / pauses up H > 25 % slower or pause-ratio sharply up	Modest effect sizes; task- and language-dependent — cross-lingual validity (Sinhala/Tamil/English) must be validated.

02 ATTENTION & LATENCY

the strongest signal for hard numbers

SIGNAL	WHAT THE SCIENCE SAYS (REFERENCE · DIRECTION · MAGNITUDE)	LOW / MEDIUM / HIGH	PRINCIPAL CAVEAT
Response latency — PVT lapse STRONG · ABSOLUTE	A "lapse" = RT > 500 ms (10-min PVT) or > 355 ms (3-min PVT-B). Lapse count & mean RT rise with fatigue/inattention.	L 0–2 lapses / bout M 3–5 lapses H > 5 lapses (calibrate to baseline)	Strong individual differences. RT $> \sim 2669$ ms $\approx$ eyes-closed microsleep.
Mean / median RT STRONG	Individual baseline. Slows under load & fatigue .	L ± 10 % M $+10$ – 25 % H $> +25$ % vs baseline	Cleanest, most objective channel; latency is measured, attention is inferred from it.

SIGNAL	WHAT THE SCIENCE SAYS (REFERENCE · DIRECTION · MAGNITUDE)	LOW / MEDIUM / HIGH	PRINCIPAL CAVEAT
PERCLOS STRONG · ABSOLUTE	% of time eyes ≥ 80 % closed, over a 1–3 min window (Wierwille 1994; FHWA/NHTSA-endorsed). ↑ with drowsiness.	L alert < 0.075 M caution 0.075–0.15 H drowsy > 0.15 (severe > 0.30)	Less reliable in real driving, aviation, older adults, moderate drowsiness. Needs eye visibility (PPE occludes).
Blink rate & duration STRONG	Fatigue → blink rate ↑ and blinks longer: duration 171–300 ms vs ~100–150 ms normal (~71–100 ms under visual load).	L near baseline M rate +20–50 % / blinks lengthening H rate sharply ↑ with prolonged closures	Opposite direction to visual cognitive-load suppression — context decides.
Head position MODERATE	Upright baseline → head droop / nod-off = drowsiness cue (driver-monitoring validated).	L stable / upright M intermittent droop H sustained droop / nodding	Confounded by ergonomics — looking down at a workpiece is not drowsiness. Corroborating cue only.
Speech (sleepiness) MODERATE	Individual baseline → slower rate, longer pauses, flatter prosody (INTERSPEECH 2011 Sleepiness, ref. Karolinska Sleepiness Scale).	L near baseline M 1–2 SD slower / flatter H > 2 SD, sustained trend	"Recovery" is a longitudinal trend, not one reading.

04 STRESS RESPONSE

weak / contested — no defensible numeric thresholds

No numeric thresholds are given — and that is the scientifically correct answer.

Pupil / blink (arousal): real but noisy acute-arousal correlates that **cannot be separated from cognitive load or luminance**. No stress cutoff.

Voice pitch (F0) / intensity: weak arousal correlates. Commercial **"voice stress analysis" / LVA for stress or deception is scientifically discredited** (chance-level); workplace emotion inference is also legally restricted (EU AI Act Art. 5(1)(f)).

Recommendation: use any voice/ocular "stress" signal only as a fatigue/arousal cue inside the safety frame — never as a stress score or lie detector. For an actual stress measure, use a validated questionnaire (e.g., PSS-10).

05 CHALLENGE–SKILL BALANCE

no biometric basis at all

There is no facial, ocular or acoustic correlate. Do not assign biometric thresholds.

Challenge–skill balance is a subjective appraisal from flow theory — measure it only with a validated instrument, the **Flow Short Scale** (Engeser & Rheinberg 2008). Likewise **Burnout** is a chronic syndrome with no momentary signature: measure with the **BAT** questionnaire (Schaufeli et al. 2020); biometrics can at most contribute a longitudinal fatigue trend.

06 CROSS-CUTTING RULES FOR DEPLOYMENT

- Baseline-first** — calibrate each person's resting values; band on deviation, not population norms.
- Multimodal fusion** — require corroboration before a "High" flag; no single channel is decisive (D'Mello & Kory 2015).
- Control luminance** for any pupil measure; **window length matters** (PERCLOS 1–3 min; PVT 3–10 min).
- PPE & ergonomics** occlude camera signals on the floor → capture at an unobstructed seated check-in; treat head-pose/gaze as corroborating, not primary.
- Cross-lingual validity** for all voice features (Sinhala / Tamil / English) must be validated, not assumed.

07 PRIMARY REFERENCES

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EU AI Act — Reg. (EU) 2024/1689, Art. 5(1)(f).

Note: The Low / Medium / High baseline-relative bands (±1 SD / 1–2 SD / > 2 SD) are an engineering convention for operationalising deviation, **not** published cutoffs. The only literature-anchored absolute cutoffs here are **PERCLOS** (0.075 / 0.15 / 0.30) and the **PVT lapse** (500 ms / 355 ms). Strength ratings reflect the published signal–construct literature, not CIQ's deployed accuracy. This is a scientific summary, not legal advice; validate all bands on your own population before deployment.